

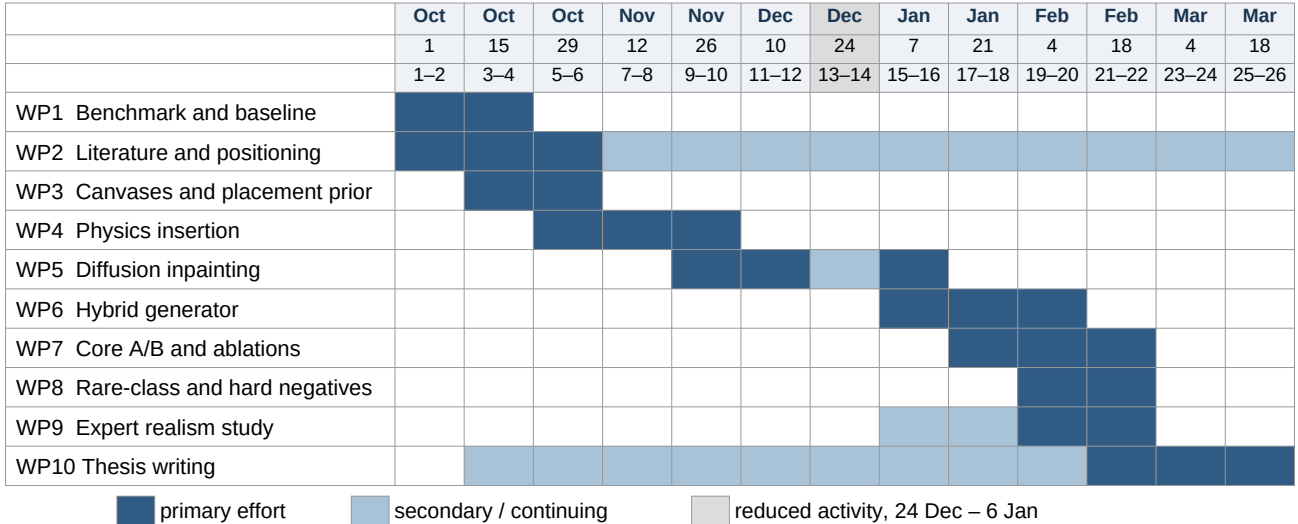
Master's Thesis — Work Plan and Timeline

Realistic Synthetic Data Generation for Defective Radiographic Testing Scans of Welds

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Start: Thursday 1 October 2026 · Submission: Wednesday 31 March 2027 · 26 weeks

1 Schedule Overview



2 Work Packages and Deliverables

Weeks	Work package	Deliverable
1–4	WP1 Benchmark. Reproduce the existing real-only segmentation baseline on SWRD. Freeze the test split, the per-class metrics and the evaluation script for the whole thesis.	Frozen benchmark; reproduced baseline (macro IoU ≈ 0.48)
1–6	WP2 Literature. Systematic survey across four communities: industrial defect synthesis, NDT simulation, medical lesion synthesis, security threat-image projection. Position against the closest 2026 weld-diffusion work.	Chapter 2 draft
3–6	WP3 Canvases. Build the pool of defect-free film regions and the seam-polygon placement prior that SWRD ships with every image.	Injection pipeline; canvas statistics
5–10	WP4 Physics insertion. Additive defect insertion in the log-attenuation domain, with the offset/gain normalisation step and attenuation-scale calibration recovered from the film itself. Exact masks by construction.	Method §3.2; first synthetic corpus; first utility numbers
9–16	WP5 Learned inpainting. Mask-conditioned diffusion inpainting fine-tuned on SWRD defect crops. Covers 16-bit grayscale handling, high-resolution tiling and a mask-fidelity loss so the generated defect stays inside its label.	Method §3.3; second synthetic corpus
15–20	WP6 Hybrid. Couple the physical constraint to the learned generator: generation in a thickness-perturbation space with a sign constraint and attenuation consistency.	Method §3.4
17–22	WP7 Core experiments. The headline A/B — identical training on physically composited versus naively blended synthetic data, scored on real film — plus the synthetic-fraction sweep and per-term ablations.	Results §4.1–4.3
19–22	WP8 Targeted studies. Rare-class injection for undercut and lack of fusion; hard-negative synthesis against the pseudo-defect class.	Results §4.4
15–22	WP9 Realism study. Blinded real-versus-synthetic study with certified RT inspectors, alongside FID/KID. Recruitment starts week 15 so inspector availability is not on the critical path.	Results §4.5
6–26	WP10 Writing. Each chapter is drafted as its work package closes, not at the end.	Full draft at week 24; final thesis at week 26

3 Milestones and Decision Gates

#	Date	Milestone / decision gate
M1	Wed 28 Oct 2026	Benchmark frozen, baseline reproduced, Chapter 2 draft submitted.
M2	Wed 9 Dec 2026	Gate: physics insertion produces usable synthetic data and a first downstream number. If the scale calibration proves unreliable, fall back to sampling perturbation amplitude from the measured real-defect distribution.
M3	Wed 20 Jan 2027	Gate: diffusion inpainting produces plausible in-mask defects. If 16-bit fine-tuning fails, drop to 8-bit tiles or a small latent model trained from scratch on SWRD.
M4	Wed 3 Feb 2027	Gate: hybrid generator running and the A/B launched. If WP6 is not viable by this date it is dropped; RQ1, RQ2 and RQ4 still complete.
M5	Wed 3 Mar 2027	All experiments finished, results frozen, expert study closed.
M6	Wed 17 Mar 2027	Complete draft to supervisor for review.
M7	Wed 31 Mar 2027	Submission.

Three of the seven milestones are explicit **decision gates** with a named fallback. The plan is built so that a failure at M3 or M4 costs a research question, not the thesis. Supervisor meetings are proposed **fortnightly**, with the four gate meetings above as the formal review points.

4 Thesis Structure and When Each Chapter Is Written

Chapter	Content	Drafted
1 Introduction	Motivation, problem statement, contributions, outline	Weeks 23–24
2 Background and Related Work	RT physics and weld defect formation; the SWRD corpus; synthetic data generation by simulation, GAN and diffusion; physically grounded insertion in NDT, medicine and security screening	Weeks 1–6, revised 23
3 Method	3.1 evaluation protocol and baseline · 3.2 physics-grounded insertion and scale calibration · 3.3 mask-conditioned diffusion inpainting · 3.4 the physics-constrained hybrid	Weeks 10, 16, 20
4 Experiments and Results	4.1 downstream utility and synthetic-fraction sweep · 4.2 the compositing-operator A/B · 4.3 ablations · 4.4 rare-class and hard-negative studies · 4.5 realism, FID/KID and the expert study	Weeks 21–23
5 Discussion	What transferred, what did not, limitations, threats to validity	Weeks 23–24
6 Conclusion and Future Work	Summary of contributions, open directions	Week 24

5 Principal Risks

Risk	Mitigation
Diffusion inpainting does not transfer to 16-bit grayscale radiographs; pretrained backbones are RGB.	Evaluated at M3. Fall back to 8-bit tiles, or a small latent model trained from scratch on SWRD. The physics track (WP4) does not depend on this and already answers RQ1 and RQ4.
Attenuation scale cannot be calibrated reliably from the film, since SWRD ships no acquisition metadata.	Evaluated at M2. The constraints that matter (log-space additivity, sign, smooth profile, noise) need no absolute scale; amplitude can instead be sampled from the measured distribution of real annotated defects.
Certified inspectors unavailable for the blinded study.	Recruitment begins week 15, seven weeks before the study window. Fallback is a reduced internal panel plus the quantitative realism metrics.
Compute contention; a single training run takes roughly eleven hours.	Runs are queued overnight and the experiment matrix is fixed at M4 so nothing is re-run on a whim.
Scope overrun on the hybrid generator (WP6).	WP6 is the designated cut. Dropping it loses RQ3's hybrid arm while leaving the physics-versus-naive A/B, which is the thesis's core claim, fully intact.

Resources are in place before week 1: the SWRD corpus and its annotations, a production training and evaluation stack with an established baseline on this exact dataset, GPU infrastructure, and access to certified RT inspectors through Deeplify GmbH. Dates follow the agreed 1 October start; the structure shifts intact if the start date moves.